

LUIS CARLOS HERNANDEZ

Director of Engineering | Software Engineering Leader | AI & Technology Strategist

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EXECUTIVE SUMMARY

Engineering leader with 20+ years of experience leading enterprise software delivery, engineering operations, and technology transformation across healthcare, insurance, financial services, aerospace, defense, transportation, retail, and eCommerce. Proven ability to lead distributed engineering teams, partner with executive stakeholders, manage vendor relationships, and deliver mission-critical platforms in complex enterprise environments.

Experienced in software delivery leadership, Agile execution, production support, release governance, architecture, cloud modernization, technical risk management, and cross-functional delivery. Combines deep hands-on engineering background with people leadership, operational discipline, and business-focused decision-making.

Currently focused on engineering operations and AI-enabled engineering intelligence, including production support transformation, executive dashboards, engineering metrics, and reusable IntelligenceOS concepts across Career Intelligence, Engineering Intelligence, and Restaurant Intelligence.

EXECUTIVE HIGHLIGHTS

- 20+ years of enterprise software engineering experience across healthcare, financial services, aerospace, defense, transportation, retail, and eCommerce.
- Led distributed engineering teams across the United States, Mexico, and India, including organizations ranging from small discovery teams to approximately 40 engineers.
- Managed delivery across enterprise clients and programs including Apple, Mastercard, Canada Life, GE Transportation, Eaton Aerospace, Crane Aerospace, Rockwell Collins, and Academic HealthPlans.
- Strong background in Java, cloud services, APIs, DevOps, Agile delivery, embedded systems, enterprise integrations, production operations, and engineering metrics.
- Builder of AI-enabled engineering intelligence concepts, dashboards, and operating models designed to improve engineering visibility, quality, and executive decision-making.

CORE COMPETENCIES

Engineering Leadership | Software Delivery | Organizational Transformation | Production Support | Release Governance | Agile Leadership | Executive Communication | Vendor Management | Cross-functional Delivery | Engineering Metrics | Operational Excellence | Cloud Transformation | Platform Engineering | AI Strategy | Java Architecture | DevOps | Risk Management | Team Development

PROFESSIONAL EXPERIENCE

Academic HealthPlans / Brown & Brown - Dallas-Fort Worth, TX | Manager, Applications Development / Software Development Manager | 2024 - Present

Lead engineering delivery and operational improvement initiatives for healthcare technology platforms supporting student health insurance operations.

- Lead production support and application development operations across distributed engineering teams, aligning day-to-day execution with business priorities, release timelines, and customer commitments.
- Drive engineering operating model improvements, including intake discipline, team capacity visibility, production support triage, release coordination, and cross-team prioritization.
- Created and evolved Engineering Intelligence dashboards and analytics concepts to improve visibility into production intake, engineering effort allocation, feature quality, operational load, and executive reporting.
- Support major platform initiatives including UC migration readiness, load testing, file ingestion validation, release planning, and operational risk reduction.
- Partner with internal teams and external vendors to clarify ownership, standardize execution expectations, improve accountability, and strengthen delivery transparency.
- Apply AI-assisted analysis to engineering operations, production support, reporting, and leadership decision-making as part of broader IntelligenceOS platform development.

Cognizant - Irving, TX | Senior Engineering Manager | 2018 - 2024

Led enterprise software engineering programs for major clients across technology, financial services, insurance, and digital commerce.

Apple - Environmental Health & Safety / Contact Gateway / Expert Admin

- Led software engineering delivery across multiple Apple business units including EHS Incidents, EHS Chemicals, EHS Inspections, EHS Industrial Hygiene, AppleCare, Apple Retail, and IS&T.

- Managed development, testing, customer release coordination, technical discussions, and operational planning across distributed teams and vendor partners.
- Led teams of approximately 13-16 engineers across onshore and offshore delivery models, improving coordination between customer stakeholders, development teams, ERM, Cority, BlueRiver, Enablon, and internal Apple organizations.
- Partnered with business and engineering stakeholders to make trade-off decisions balancing business goals, platform strategy, customer experience, risk, and maintainability.
- Communicated progress, escalated risks, removed blockers, and improved execution visibility through metrics, documentation, and delivery governance.

Mastercard - Mastercard Send Partner Program

- Managed engineering delivery for a distributed organization of approximately 40 software engineers in Mexico across four Scrum teams.
- Supported the Mastercard Send Partner Program by coordinating complex technical design, delivery execution, customer needs, and platform maintainability.
- Partnered with software engineers, principals, technical program managers, customers, and technology providers to simplify designs and align delivery with long-term operational needs.
- Led delivery using Java 11, Oracle, Spring, REST/SOAP web services, JSON, JUnit, Maven, Angular, Tomcat, Redis, and Swagger.

Canada Life - Mobile App Discovery

- Led discovery for an Android/iOS insurance application with a cross-functional team of five.
- Coordinated stakeholder interviews across DevOps, Testing, Development, Design, API, Product Ownership, and Architecture.
- Produced roadmap and project planning outputs focused on mobile experience redesign, personalization, contextual engagement, Salesforce Marketing Cloud integration, and geofencing capabilities.

Digital Room Inc. - Arlington, TX | Manager, Software Engineering | 2020 - 2021

Led multiple technical teams supporting large-scale eCommerce and production workflow platforms for the printing industry.

- Managed approximately 10 engineers across AWS-based enterprise eCommerce and Android workflow systems.
- Led technical direction, Agile delivery, customer issue resolution, stakeholder coordination, and production support.
- Participated in product, architecture, strategy, and executive planning discussions shaping customer-facing software capabilities.
- Designed and delivered cloud-based capabilities using AWS Lambda, S3, DynamoDB, DocumentDB, Aurora, EC2/ECS, OAuth2, REST APIs, and CloudWatch.
- Designed a Watchdog self-health monitoring mechanism integrated with AWS CloudWatch and standardized health visibility across applications.

Leonardo DRS - Richardson, TX | Manager, Software Engineering | 2018 - 2020

Led embedded software engineering teams supporting defense and aerospace systems.

- Led design and delivery of embedded software systems in partnership with hardware, systems engineering, and program teams.
- Managed schedules, staffing, resource planning, risk, requirements, software design documentation, and qualification testing.
- Oversaw Software Development Plans, Software Requirements Specifications, Software Description Documents, and Software Test Plans.
- Partnered with Line of Business leadership and program teams on resource planning, product roadmaps, proposals, and engineering Basis of Estimates.
- Mentored engineering staff and supported customer communication, technical reviews, and compliance with quality standards.

General Electric Transportation - Fort Worth, TX | Technical Lead Software Engineer | 2015 - 2018

- Led technical operations for system requirements, software design, implementation, and testing of locomotive Engine Control Unit applications using Java.
- Coordinated with hardware, systems, vendor, and engineering teams to align design and implementation decisions.
- Designed and implemented seven Java-based modules including CMS, Onboard Maintenance, Speed Module, GPS, CAB, EXC, and AUX modules.
- Supported migration of legacy software to Predix-based architecture using embedded Tomcat, Spring, Hypervisor, and Hibernate.
- Built test harnesses using Java, Python, and shell scripting.

Eaton Aerospace - Fort Worth, TX | Lead Software Engineer | 2010 - 2015

- Led software architecture, requirements, interface design, and implementation in C, C++, and Java for aerospace test bench and embedded software systems.
- Managed and developed staff supporting software development and formal qualification of aerospace test bench systems.
- Implemented aircraft systems simulation using Java/J2EE, Linux, Apache Tomcat, Python scripts, and Java applications.
- Supported software planning, requirements, test plans, customer communication, and qualification activities.

Earlier Engineering Experience | Crane Aerospace | Astronautics of America | Rockwell Collins | Lockheed Martin | Atmos Energy | U.S. Army / 82nd Airborne Division | 2001 - 2010

- Delivered Java/J2EE, C/C++, Python, Perl, Hibernate, JDBC, Tomcat, Linux, MySQL, and real-time/embedded software solutions across aerospace, defense, gas utilities, and enterprise systems.
- Supported aviation and military systems including A400M cargo aircraft simulation, CMS Maintenance System, anti-lock braking simulation, test harnesses, and certification support systems.
- Spent six months onsite in Madrid, Spain supporting customer integration for aerospace systems.
- Served as Weapons Sergeant in the U.S. Army / 82nd Airborne Division, building leadership, discipline, mission focus, and operational accountability.

AI & ENGINEERING INNOVATION

Founder and architect of IntelligenceOS, a modular AI-powered enterprise intelligence platform concept designed to support multiple business domains through reusable plugins, normalized data, knowledge graphs, AI recommendations, and executive dashboards.

- Career Intelligence - AI-powered career platform for resume optimization, job posting analysis, interview preparation, recruiter tracking, and application analytics.
- Engineering Intelligence - AI-assisted engineering operations platform for production support analytics, effort allocation, quality visibility, intake attribution, and executive dashboards.
- Restaurant Intelligence - AI operating system concept for restaurants and food processing companies covering orders, inventory, accounting, production, kitchen operations, drivers, customers, marketing, and camera analytics.

TECHNICAL LEADERSHIP

Java | Spring | REST APIs | SOAP | JSON | JUnit | Maven | Angular | Tomcat | Redis | Swagger | AWS Lambda | S3 | DynamoDB | DocumentDB | Aurora | EC2/ECS | CloudWatch | OAuth2 | Git | GitLab | Jenkins | Jira | SQL | Python | Shell Scripting | C/C++ | Embedded Systems | Linux | Unix | DevOps | Agile | CI/CD | Production Support | Architecture | Engineering Metrics

EDUCATION

Iowa State University - Ames, Iowa | Master of Science, Computer Engineering

Campbell University - Buies Creek, North Carolina | Bachelor of Science, Computer Science

University of Texas at El Paso - El Paso, Texas | Computer Science coursework prior to transfer